

### Experiment 7.1.2: Possible Combinations from Three Digit

1. Aim To design and implement a Python program that accepts three single-digit inputs (separated by spaces) and prints all six possible 3-digit combinations of these digits. Each digit should be used exactly once in each combination. The program maintains leading zeros if they are present

Algorithm : 1 **Start**

#### 2 **Input three digits**

- Read three space-separated digits: a, b, c

#### 3 **Store digits**

- Assign them to variables:  
a = first digit  
b = second digit  
c = third digit

#### 4 **Generate all permutations**

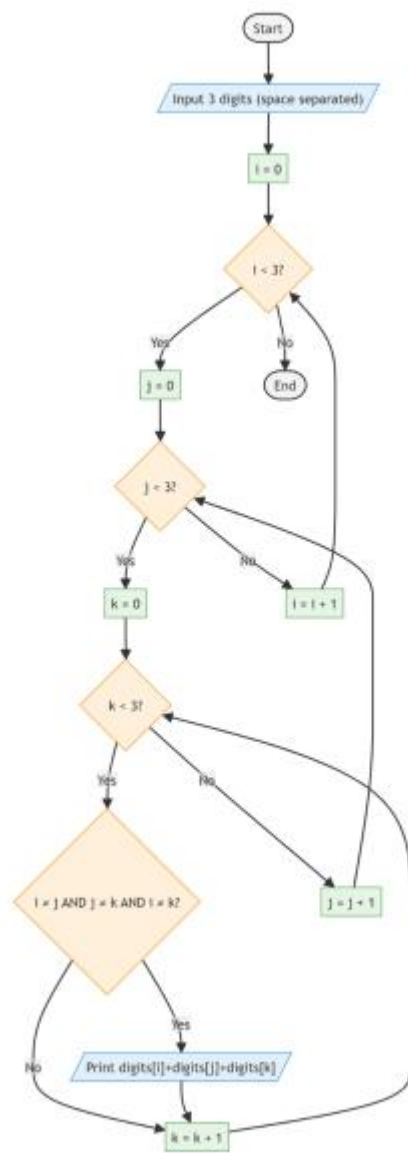
- Form all possible arrangements:
  - a b c
  - a c b
  - b a c
  - b c a
  - c a b
  - c b a

#### 5 **Print combinations**

- For each arrangement:
  - Concatenate digits as a string (to preserve leading zeros)
  - Print the result

#### 6 **End**

**Flowchart :**



## Program :

CODETANTRA

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7.1.2. Possible Combinations from Three Digits

Write a Python program to accept three digits and print all possible 3-digit combinations that can be formed using those three digits. Each digit should be used exactly once in each combination.

For three digits  $a$ ,  $b$ , and  $c$ , there are 6 possible arrangements based on their indices:  $abc$ ,  $acb$ ,  $bac$ ,  $bca$ ,  $cab$ ,  $cba$ .

**Input Format:**

- Single line contains three space-separated non-negative integers,  $a$ ,  $b$ , and  $c$ , representing the three digits

**Output Format:**

- Print all possible 3-digit combinations
- Each combination should be printed as a 3-digit number on a separate line
- Print combinations as they appear (maintain leading zeros if first digit is 0)

**Constraints:**

- $0 \leq$  each digit  $\leq 9$
- Digits can be same or different

Sample Test Cases

combinat...

```
1
2 digits = input().split()
3 a, b, c = digits
4 combinations = [
5     a + b + c,
6     a + c + b,
7     b + a + c,
8     b + c + a,
9     c + a + b,
10    c + b + a
11 ]
12 for combo in combinations:
13     print(combo)
```

Average time0.007 s7.29 msMaximum time0.011 s11.00 ms

3 out of 3 shown test case(s) passed4 out of 4 hidden test case(s) passed

Test case 111msDebug

| Expected output | Actual output |
|-----------------|---------------|
| 1 2 3           | 1 2 3         |
| 123             | 123           |
| 132             | 132           |
| 213             | 213           |
| 231             | 231           |
| 312             | 312           |

TerminalTest cases

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